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GitHub Link	https://github.com/YOGAMADHA/MLA0405-DEEP-LEARNING

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3
Comparative Analysis

COURSE NAME:DEEP LEARNING
SLOT :D

COURSE CODE:MLA0405

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Name of the student: Yoga Madha

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05

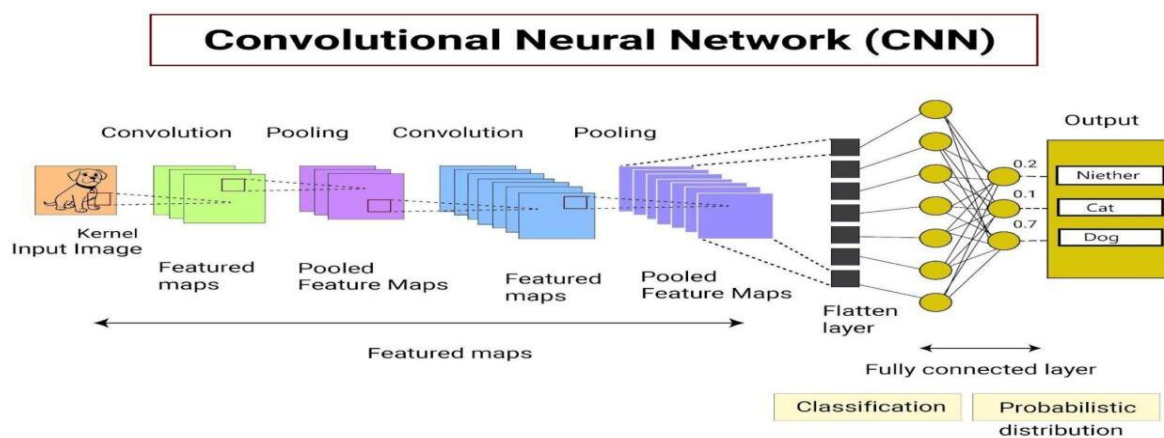
Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Answer:

A medical imaging organization wants to classify X-ray images. A **Convolutional Neural Network (CNN)** is generally more suitable than a traditional **Fully Connected Neural Network (FCNN)** because CNNs are specifically designed to preserve and learn spatial relationships in image



Comparison

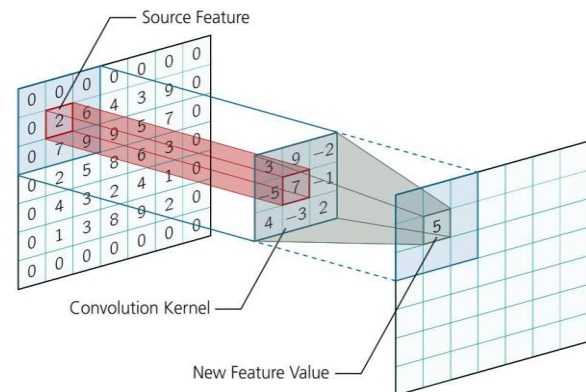
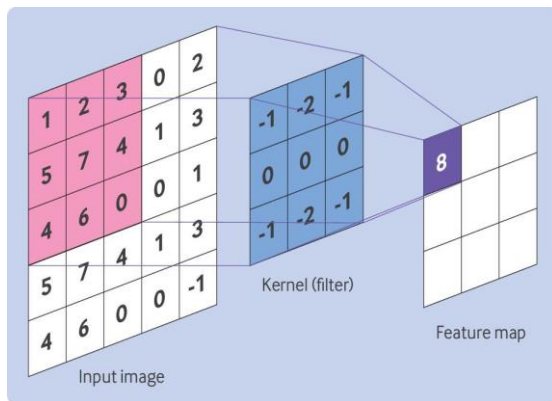
Parameter	CNN	Fully Connected Neural Network
Architectural design	Convolution, activation, pooling and dense layers	Mainly fully connected layers
Input representation	Preserves image height, width and channels	Usually flattens image into a vector
Motivation	Designed to exploit spatial relationships	General-purpose learning architecture
Spatial feature extraction	Excellent; learns local and hierarchical features	Limited because spatial arrangement is not explicitly preserved
Parameters	Relatively fewer because of local connections and shared weights	Very large for high-resolution images
Computational complexity	Efficient for image processing	Becomes expensive as image size increases
Feature learning	Automatically learns edges, textures and shapes	Learns features through dense connections
Image handling	Highly suitable	Less suitable

Role of Convolutional Layers

The convolutional layer is the main feature-extraction component of a CNN. It applies small learnable filters to local regions of an image.

For example, a 3×3 filter can scan across an X-ray and detect particular patterns.

Image $\rightarrow$ Filter $\rightarrow$ Feature Map



Early convolutional layers can detect simple features such as:

- Edges
- Lines
- Corners
- Basic textures

Deeper layers combine these features into more complex structures that can be useful for identifying abnormalities in X-rays.

Role of Filters

A filter or kernel contains learnable weights. Different filters learn different features. One filter may respond strongly to horizontal edges while another may learn textures or boundaries.

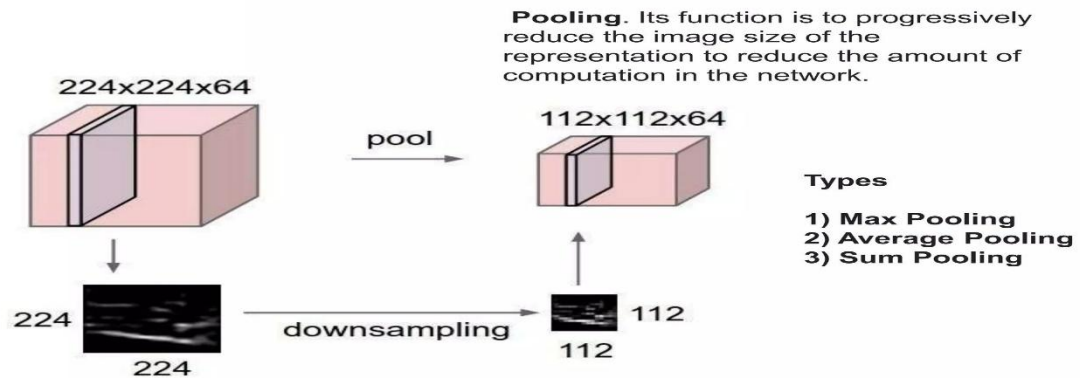
Because the same filter is reused across the entire image, CNNs require far fewer parameters than fully connected networks.

Role of Pooling

Pooling reduces the spatial dimensions of feature maps while retaining important information.

For example, 2×2 **max pooling** selects the maximum value from each small region.

Pooling process



Pooling helps to:

- Reduce computational requirements.
- Reduce the size of feature maps.
- Retain important features.
- Provide some tolerance to small image shifts.

Parameter Comparison

Suppose a grayscale X-ray has a resolution of 224×224 :

$$224 \times 224 = 50,176 \text{ pixels.}$$

If these pixels are directly connected to 1,000 neurons:

$$50,176 \times 1,000 = 50,176,000$$

weights are required.

A CNN using a 3×3 filter requires only:

$$3 \times 3 = 9$$

weights per filter, with the same weights reused across many image locations.

Justification

CNNs are more suitable because they:

- Preserve spatial information.
- Automatically learn visual features.
- Use parameter sharing.
- Require fewer parameters.
- Reduce computational requirements.
- Handle high-resolution images more efficiently.
- Learn hierarchical features useful for medical image classification.

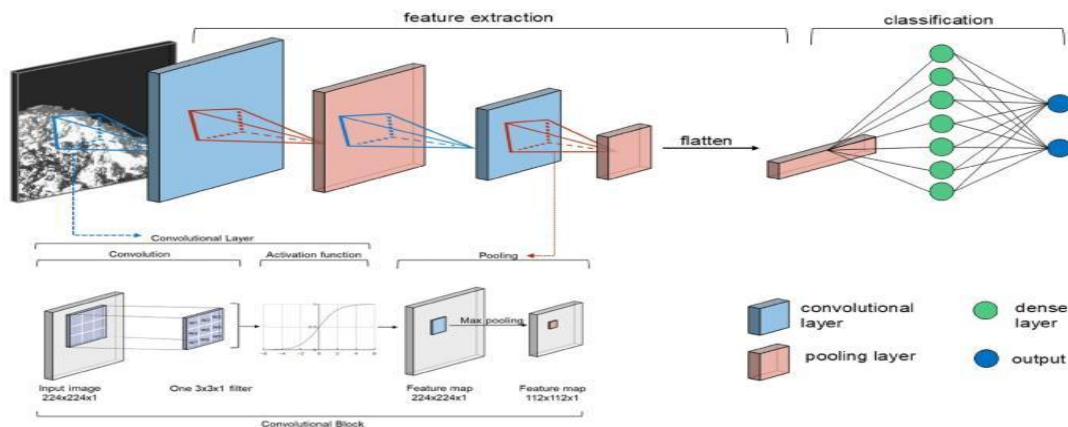
Conclusion: For X-ray classification, CNNs are clearly preferable because their architecture matches the spatial nature of medical images.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers**, **pooling layers**, **activation layers**, and **fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters** and **deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Answer:

In industrial defect detection, different CNN layers perform different but complementary functions. Convolutional layers extract features, pooling layers reduce spatial dimensions, activation layers introduce non-linearity, and fully connected layers perform final classification.



Comparison of CNN Layers

Layer	Main Function	Feature Extraction	Spatial Information	Computational Requirement	Classification Role
Convolutional	Detect features	Very high	Preserves local spatial information	Moderate	Indirect
Pooling	Reduce dimensions	Selects important information	Reduces spatial size	Low	Indirect
Activation	Introduce non-linearity	Enables complex feature learning	Usually preserves dimensions	Low–moderate	Indirect
Fully Connected	Combine learned features	Uses extracted features	Spatial structure mostly lost	High if large	Direct

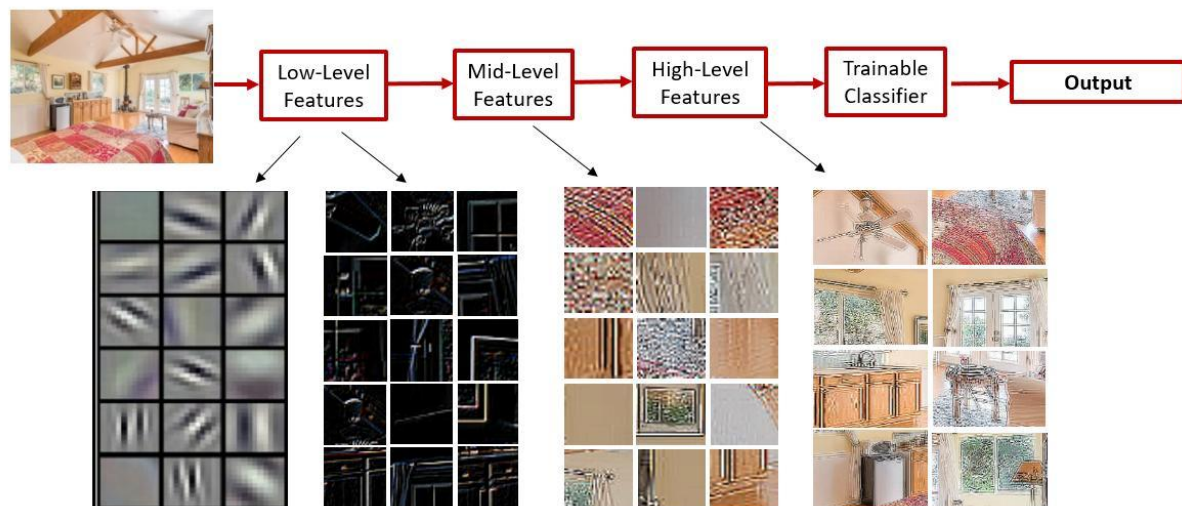
1. Convolutional Layers

Convolutional layers apply filters to the input image.

In defect detection, they can learn:

- Scratches
- Cracks
- Holes
- Surface irregularities
- Broken edges
- Texture variations

▪ Input image pixels → Edges → Textures → Parts → Objects



2. Pooling Layers

Pooling reduces feature-map dimensions.

For example:

$$\begin{array}{c}
 4 \times 4 \text{ Feature Map} \\
 \downarrow \\
 2 \times 2 \text{ Max Pooling} \\
 \downarrow \\
 2 \times 2 \text{ Feature Map}
 \end{array}$$

This reduces computation and makes the representation less sensitive to small positional changes.

3. Activation Layers

Activation functions introduce non-linearity into the network.

A commonly used activation is **ReLU**:

$$\text{ReLU}(x) = \max(0, x)$$

Without activation functions, multiple neural-network layers would behave approximately like a single linear transformation.

ReLU allows CNNs to learn complex defect patterns.

Early-Layer vs Deeper-Layer Filters

Aspect	Early-Layer Filters	Deeper-Layer Filters
Features learned	Edges, lines and corners	Complex shapes and patterns
Receptive field	Small	Larger
Feature complexity	Simple	Complex
Example in manufacturing	Surface edges	Crack or defect pattern
Specificity	General	More task-specific
Spatial information	Strongly preserved	More abstract

Critical Evaluation

Using only convolutional layers would preserve too much spatial information and increase computational requirements. Pooling reduces this burden, while activation functions allow complex relationships to be learned. Fully connected layers then convert the learned high-level representation into a classification decision.

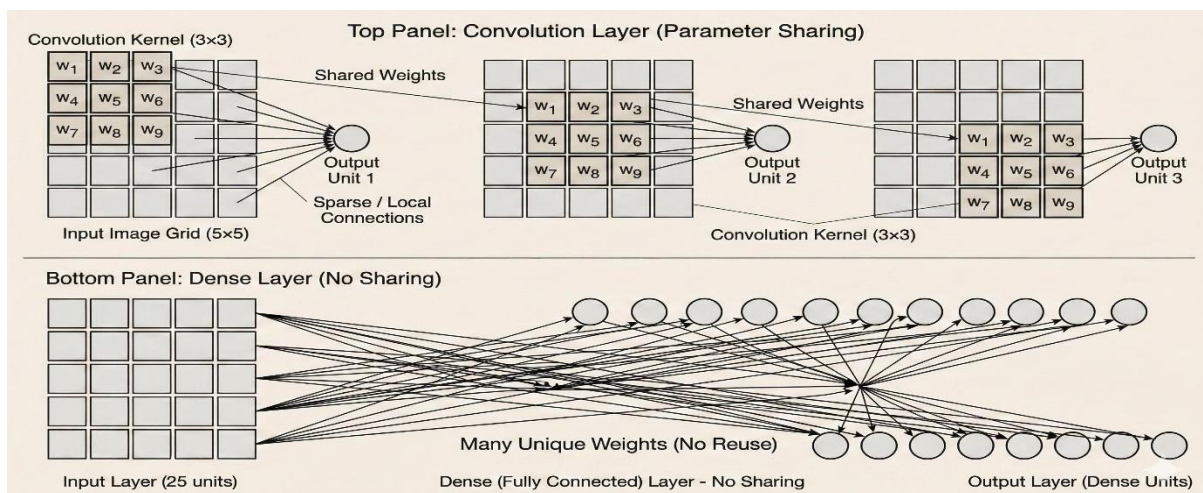
Conclusion: The layers work together rather than independently. Convolution extracts features, pooling compresses them, activation enables non-linear learning, and fully connected layers classify the product. Early filters detect basic patterns while deeper filters detect meaningful defect structures.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Answer:

A surveillance system processing thousands of high-resolution images requires efficient learning. CNN parameter sharing provides a major advantage over independent weights in fully connected networks.



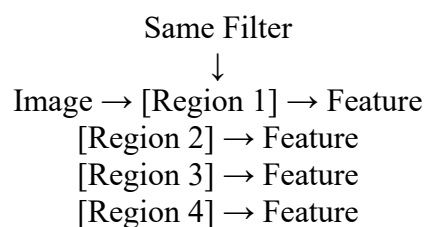
Comparison

Parameter	CNN Parameter Sharing	Fully Connected Independent Weights
Weights	Same filter reused across image	Different weight for each connection
Number of parameters	Relatively small	Very large
Memory requirement	Lower	Higher
Computation	More efficient for images	Expensive for high-resolution images
Feature detection	Same feature can be detected anywhere	Feature-location relationship must be learned separately
Translation handling	Better tolerance to shifts	Poorer
Learning efficiency	High	Lower
Scalability	Better for large image datasets	Difficult for very large images

Parameter Sharing

Suppose a CNN has a 3×3 filter: $3 \times 3 = 9$ weights.

Instead of learning a different set of nine weights for every image location, the CNN **reuses the same nine weights** across the entire image.



The same filter can therefore detect a particular feature wherever it occurs.

Independent Weights

In a fully connected network, each input pixel has independent connections to neurons.

For a high-resolution image, this can create millions of weights.

Memory and Computational Complexity

Parameter sharing substantially reduces:

- Model size
- Memory consumption
- Number of computations

- Training time
- Storage requirements

This is particularly important when processing thousands of high-resolution surveillance images.

Translation-Related Invariance

Suppose a surveillance camera detects a person's face.

If the face appears slightly to the left or right, a shared convolutional filter can still detect similar local patterns.

Therefore, CNNs provide **translation-related robustness**.

Strictly speaking, convolution itself provides **translation equivariance**, while pooling and later processing can provide a degree of **translation invariance**.

Critical Evaluation

Parameter sharing does not mean CNNs are computationally free. Large CNNs can still require significant GPU resources. However, compared with a fully connected approach, weight sharing dramatically reduces the number of learnable parameters and makes large-scale image processing practical.

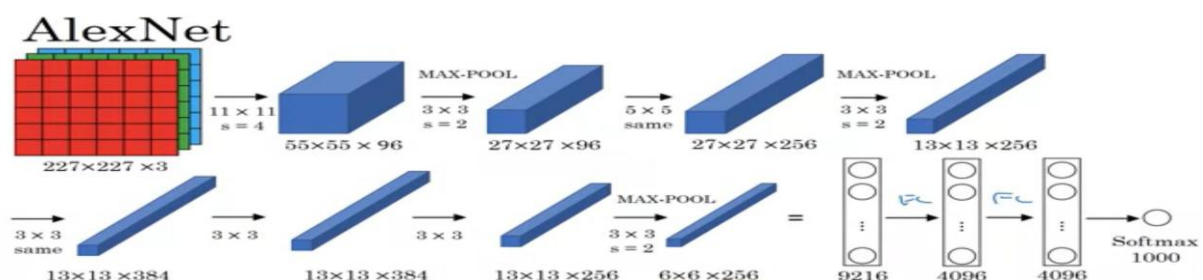
Conclusion: Parameter sharing is one of the fundamental reasons CNNs are successful for image processing. It allows the same visual feature detector to be reused across the image, reducing memory and computation while improving learning efficiency.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet** and **ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Answer:

AlexNet and ResNet are important CNN architectures, but they represent different stages in the development of deep learning. AlexNet demonstrated the effectiveness of deep CNNs for large-scale image classification, while ResNet introduced **residual connections** that made much deeper networks easier to train.



Comparison

Parameter	AlexNet	ResNet
Architecture	Conventional deep CNN	Residual CNN
Depth	About 8 learned layers	Common versions include 18, 34, 50, 101 and 152 layers
Layer organization	Convolution + pooling + fully connected	Convolution + residual blocks
Filters	Learns hierarchical visual features	Learns hierarchical features through residual blocks
Parameters	Relatively large, especially dense layers	Depends on variant; deeper models can still be efficient
Feature extraction	Good	Excellent for complex visual tasks
Training difficulty	Easier than very deep networks but limited in depth	Residual connections make deep training easier
Vanishing gradients	More susceptible as depth increases	Skip connections help reduce the problem
Computational cost	Moderate	Can be high for deep variants
Classification capability	Strong for its time	Generally stronger for complex modern tasks
Complex image recognition	Limited compared with modern deep architectures	Highly suitable

AlexNet

AlexNet consists of convolutional layers followed by pooling and fully connected layers.

It became particularly influential because it demonstrated that deep CNNs trained with GPUs could achieve excellent image-classification performance.

Its architecture includes:

Input → Convolution → ReLU → Pooling → Convolution → Pooling → Fully Connected → Output

Vanishing-Gradient Problem

As neural networks become deeper, gradients can become extremely small during backpropagation. This makes it difficult for earlier layers to learn.

ResNet addresses this using **skip connections**.

Instead of forcing a block to directly learn:

$$H(x)$$

it learns a residual function:

$$F(x)=H(x)-x$$

and produces:

$$H(x)=F(x)+x$$

This makes optimization of deep networks easier.

Critical Evaluation

AlexNet is historically important and can still be useful for simpler applications or educational purposes. However, ResNet provides much better scalability for complex recognition tasks because its residual architecture enables considerably deeper networks.

Recommendation

For a **complex image recognition application, ResNet is the better choice.**

Reasons include:

- Greater depth.
- Better hierarchical feature extraction.
- Improved training of deep networks.
- Reduced impact of vanishing gradients.
- Strong performance on complex visual patterns.
- Availability of different model sizes such as ResNet-18, ResNet-34 and ResNet-50.

Conclusion: AlexNet is an important foundational CNN architecture, but ResNet is more suitable for modern complex image recognition because residual connections allow deeper and more effective feature learning.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Answer:

The agricultural organization observes very high training accuracy but poor performance on unseen leaf images. This is a typical indication of **overfitting**.

A CNN without suitable regularization can memorize training images instead of learning general disease patterns.

Comparison

Parameter	CNN Without Regularization	CNN With Regularization
Training accuracy	Often very high	High but more controlled
Test accuracy	May be poor	Usually better
Overfitting	High risk	Reduced
Generalization	Poor	Better
Training stability	Can be unstable	Batch normalization improves stability

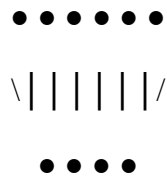
Parameter	CNN Without Regularization	CNN With Regularization
Data diversity	Limited to training dataset	Increased through augmentation
Model complexity	May memorize training patterns	Better constrained
Computational requirements	Lower initially	Slightly higher
Real-world performance	Often unreliable	More reliable

1. Dropout

Dropout randomly disables a proportion of neurons during training.

For example:

Before Dropout:



After Dropout:



This prevents the network from depending too strongly on particular neurons.

Advantage

Dropout reduces overfitting and encourages the network to learn more robust representations.

Limitation

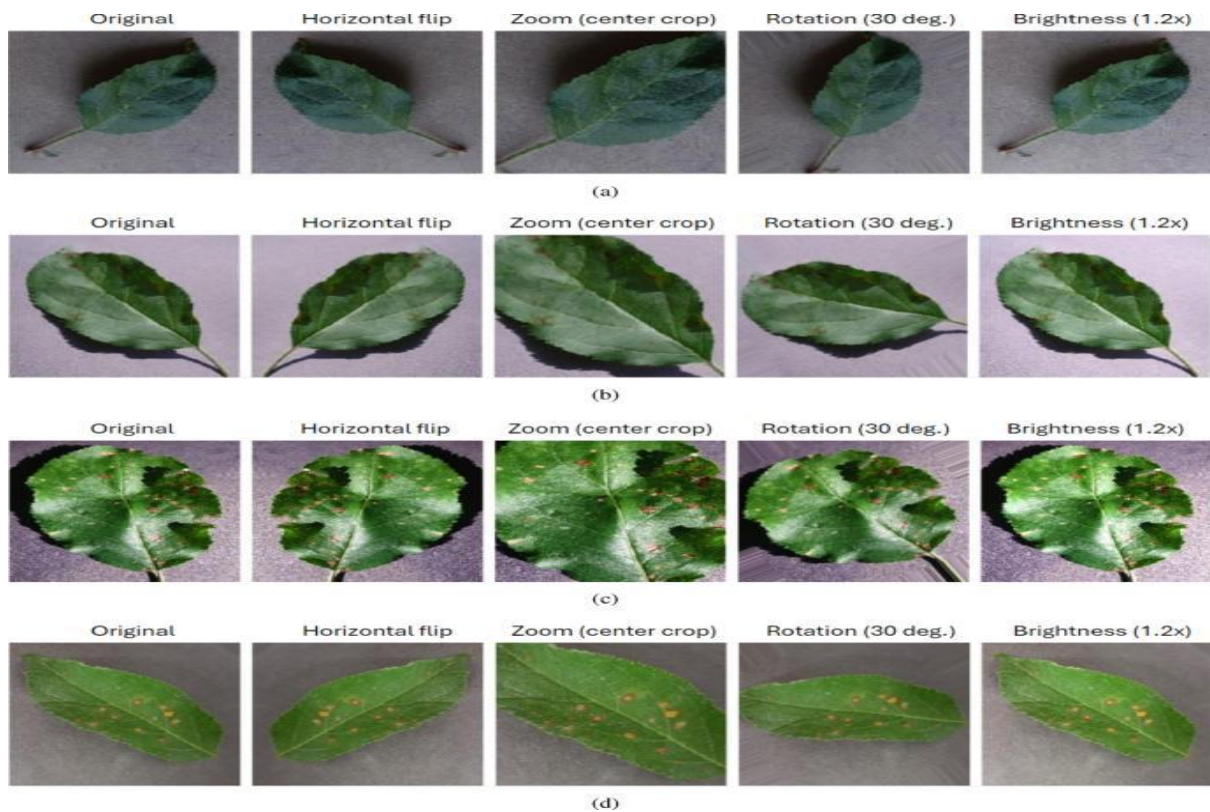
Too much dropout can cause underfitting and slow learning.

2. Data Augmentation

Data augmentation creates modified versions of existing leaf images.

Examples include:

- Rotation
- Horizontal flipping
- Small translations
- Zooming
- Cropping
- Brightness variation
- Small scaling changes



For example:

Original Leaf → Rotation → Flip → Zoom → Brightness Change

This gives the CNN more variation during training.

Advantage

Data augmentation is especially useful because real agricultural images may have different:

- Camera angles
- Lighting conditions
- Leaf orientations
- Backgrounds
- Distances

Therefore, augmentation can significantly improve generalization.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training.

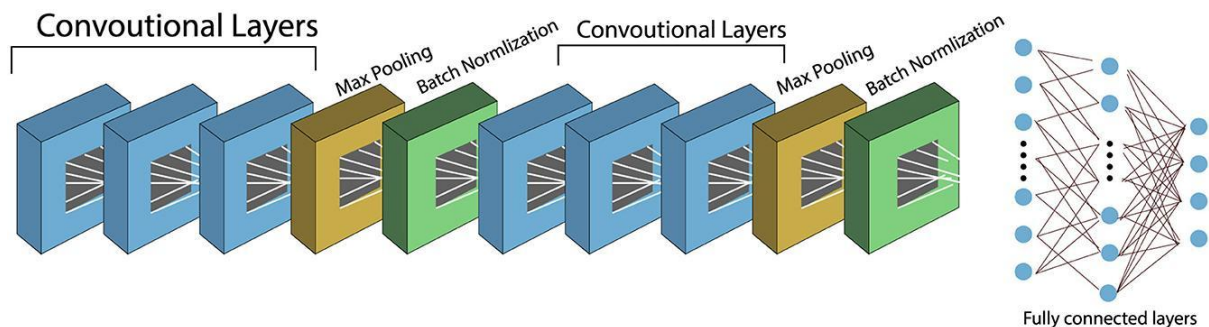
Conceptually:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

where (μ) is the batch mean and (σ^2) is the batch variance.

It can:

- Improve training stability.
- Allow effective training with suitable learning rates.
- Help gradients behave more consistently.
- Often speed up convergence.



Comparison of the Three Techniques

Technique	Main Purpose	Effect on Overfitting	Main Benefit	Possible Limitation
Dropout	Randomly deactivate neurons	Reduces overfitting	Robust feature learning	Too much can cause underfitting
Data Augmentation	Increase training variation	Strongly reduces overfitting	Better real-world generalization	Adds preprocessing/training time
Batch Normalization	Normalize activations	Can provide regularization effect	More stable and efficient training	Adds computation and layers

Critical Evaluation

Using only dropout may not be sufficient if the dataset contains too few leaf images. Similarly, batch normalization primarily improves training behavior rather than solving every source of overfitting.

For plant disease classification, **data augmentation is particularly important** because the appearance of leaves can naturally vary due to orientation, lighting, camera position, and scale.

A suitable approach is:

Leaf Images → Data Augmentation → Convolution → Batch Normalization → Activation → Pooling → Dropout → Fully Connected → Disease Classification

Recommended Approach

The organization should use a **regularized CNN combining data augmentation, batch normalization, and appropriately selected dropout.**

The priority should be:

1. Data Augmentation

Increase the diversity of training examples.

2. Batch Normalization

Improve training stability and convergence.

3. Dropout

Reduce excessive dependence on individual neurons and further control overfitting.

The exact dropout rate should be tuned using validation performance rather than choosing an unnecessarily large value.

Conclusion

A non-regularized CNN may achieve excellent training accuracy while failing on unseen leaf images because of overfitting. A properly regularized CNN learns more general disease characteristics and is therefore more reliable in real agricultural environments.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand